A3-PS015-TC233

SignalSync: Explainable Home and Last-Mile Connectivity Diagnosis System

Shubha M
shubham.ec25@bmsce.ac.in

AIORI Hackathon 3

October 14, 2026

Problem Description

- Home internet users often experience weak Wi-Fi signals, slow connections, and frequent disconnections.
- Users may struggle to identify the cause of these connectivity problems.
- Troubleshooting without clear information can be time-consuming, especially for non-technical users.
- There is a need for an affordable, user-friendly system that helps identify possible network issues and suggests solutions.

Background

Wi-Fi connectivity plays an important role in everyday activities such as online learning, communication, and accessing digital services. However, users may experience connectivity problems such as weak signal strength, unstable connections, and frequent disconnections. Identifying the exact cause of these issues can be difficult without proper network information.

Wi-Fi signal strength and connectivity status provide useful information for understanding network performance. Analysing these measurements can help identify possible connectivity issues and guide users towards appropriate troubleshooting steps.

SignalSync aims to simplify this process by using an ESP32 to collect Wi-Fi information and display the results through a user-friendly dashboard, helping users understand possible network problems and their solutions.

Literature & RFC Review

Existing Wi-Fi troubleshooting methods use network measurements such as signal strength and connectivity status to identify possible connectivity issues. However, users without technical knowledge may find it difficult to interpret these measurements and determine the cause of network problems.
The ESP32 provides built-in Wi-Fi capabilities that can be used to monitor signal strength and connection status. Networking standards such as ICMP also support connectivity diagnostics. SignalSync combines these technologies with a user-friendly dashboard to analyse network information, explain possible causes of connectivity issues, and recommend suitable troubleshooting steps.

.

Proposed Solution

SignalSync uses an ESP32 to collect Wi-Fi information, including signal strength and connectivity status. The collected data is sent to a user-friendly dashboard for analysis and display.
The system identifies possible connectivity issues, explains their likely causes, and recommends suitable troubleshooting steps. A fault simulator will also demonstrate common network problems during the project presentation.
By combining electronics and software, SignalSync aims to make home internet troubleshooting simpler and more accessible to users.

Optimization Proposed

Before SignalSync	After SignalSync
Users struggle to identify network issues.	The dashboard displays network information.
Wi-Fi signal strength is difficult for users to interpret.	Signal strength is presented in an understandable format.
Troubleshooting is often confusing.	The system suggests possible causes and solutions.
Users may not know where to start.	Guided troubleshooting helps users take the next step.

Solution Architecture & Design

Visual: Flowchart or diagram showing the optimization process.



Figure: System Architecture UML Diagram

Data Flow Diagram

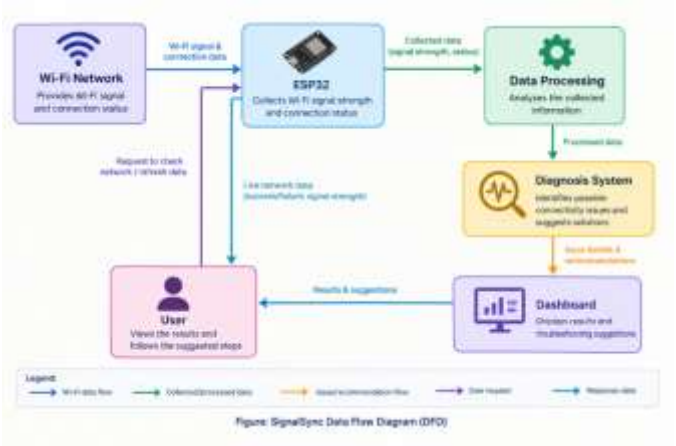


Figure: Data Flow Diagram (DFD)

Deployment Timeline

The SignalSync project will be developed in stages, starting with requirement analysis and system design. Next, the ESP32 will be configured to collect Wi-Fi information, followed by the development of a dashboard for displaying network data. The diagnosis logic and fault simulator will then be integrated and tested. Finally, the complete prototype will be demonstrated and evaluated.

Milestone

Phase	Activity
Phase 1	Problem analysis and system design
Phase 2	ESP32 setup and Wi-Fi data collection
Phase 3	Dashboard development
Phase 4	Diagnosis logic and fault simulation
Phase 5	Integration, testing, and demonstration

Bibliography I

1. Espressif Systems, *ESP-IDF Programming Guide: Wi-Fi API*. Official ESP32 Wi-Fi documentation.

2. J. Postel, *Internet Control Message Protocol (ICMP)*, RFC 792, 1981.

3. P. Mockapetris, *Domain Names—Implementation and Specification*, RFC 1035, 1987.

Thank You for Your Attention!